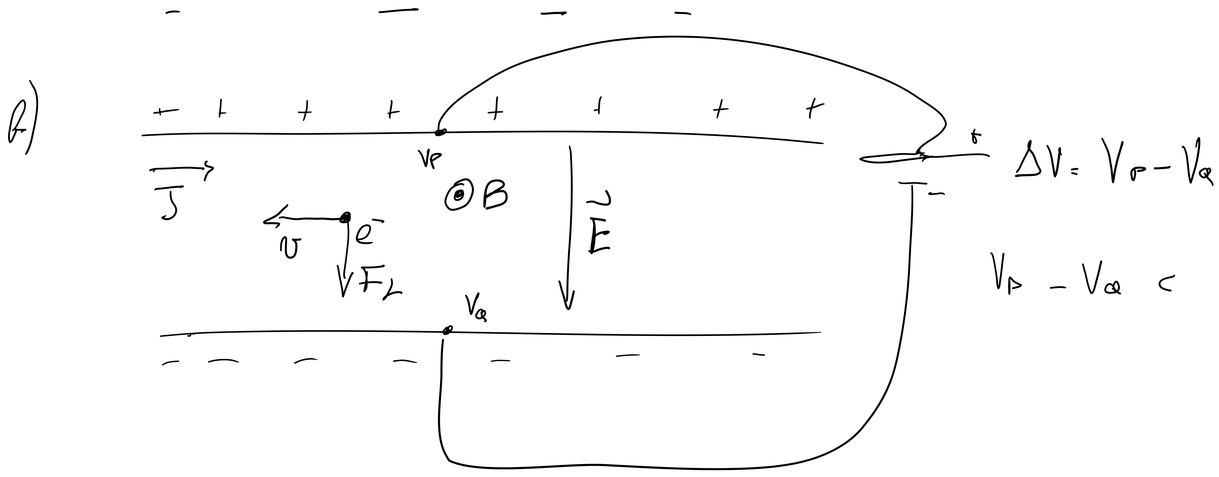
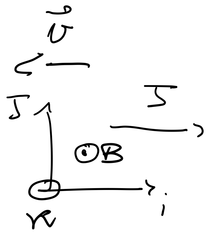
Effetto hall

2/5/21 11:13 AM



All'equilibrio

$$\vec{F}_e + \vec{F}_L = 0$$



$$-e\vec{E} + (-e)\vec{v} \times \vec{B} = 0$$

$$\vec{E} = -\vec{v} \times B \qquad \vec{J} := n q \vec{v}$$

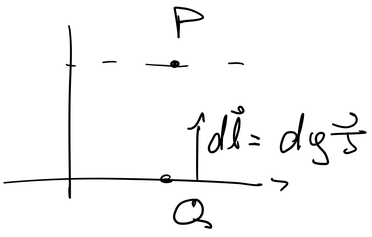
$$\vec{E} = \frac{1}{en}\vec{J} \times \vec{B} \qquad \vec{J} := (-e) n \vec{v}$$

$$\vec{E} = \frac{1}{en}|J||B|\, i \times K \qquad \frac{\vec{J}}{(-e)n} = \vec{v}$$

$i \times K =$

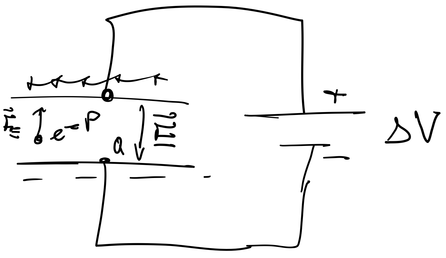
$$\vec{E} = -\frac{|J||B|}{en}\vec{J}$$ Verso il Basso

$$V(P) - V(Q) = -\int_Q^P \vec{E} \cdot d\vec{l}$$



$$V(P) - V(Q) = -\int_0^d \frac{-|J||B|}{en}\vec{J} \cdot \vec{J}\, dy$$

$$V(P) - V(Q) = \frac{|J||B|}{en} d = \Delta V$$



NB

$$V(P) - V(Q) > 0$$

NB $V(P) > V(Q)$

Lastra positiva Lastra negativa

Gli e^- si spostano spontaneamente da zone a potenziale minore a maggiore

$$\Delta V = \frac{|J||B|}{en} d$$

APPLICAZIONI:

① Determinare il segno dei portatori

② Misura di B